

A. Mathematical Formulation

Prior: $P(H) = 0.50$

Evidence FOR:

• PHQ-9 = 18	0.90
• Anhedonia	0.85
• Sleep disturbance	0.80
• Duration > 2 weeks	0.95

Evidence AGAINST:

• Recent bereavement	0.60
• Thyroid dysfunction	0.40

Likelihood Ratios:

LR_for = 2.45
LR_against = 0.68
LR_combined = 1.67

Posterior: $P(H|E) = 0.73$

B. Turbulance Execution Output

Creating Resolution:

"patient has major depressive disorder"

Evidence FOR:

- ✓ "PHQ-9 score = 18" (0.90)
- ✓ "anhedonia present" (0.85)
- ✓ "sleep disturbance" (0.80)
- ✓ "duration > 2 weeks" (0.95)

Evidence AGAINST:

- ✗ "recent bereavement" (0.60)
- ✗ "thyroid dysfunction" (0.40)

Bayesian Update:

Posterior certainty: 0.73

Perturbation Validation:

Robustness score: 0.87 (high)

C. Clinical Application

Diagnosis: Major Depressive Disorder

Confidence: 73%

Robustness Analysis:

Remove PHQ-9		$\Delta = -0.05$
Remove anhedonia		$\Delta = -0.03$
Remove sleep		$\Delta = -0.02$
Remove duration		$\Delta = -0.08$

Most Critical Evidence:

- Duration > 2 weeks ($\Delta = -8\%$)
- PHQ-9 score ($\Delta = -5\%$)

Recommendation:

Proceed with treatment protocol
(high confidence, robust evidence)